Observer Pattern

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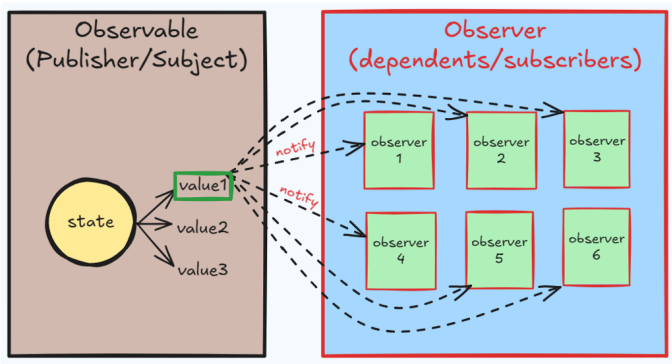
Resources

- Video link → 41. All Behavioral Design Patterns | Strategy, Observer, State, Template, Command, Visitor, Memento
- Video link → 3. Observer Design Pattern Explanation (Hindi) | Design Interview Question | LLD System Design

Definition

> ***The Observer Pattern is a behavioral design pattern where an object (aka the "subject" or "observable" or “publisher”) maintains a list of dependents (called "observers") and automatically notifies them whenever there is a change in its state. The pattern also allows addition and removal of observers at runtime.***

Concept && Coding

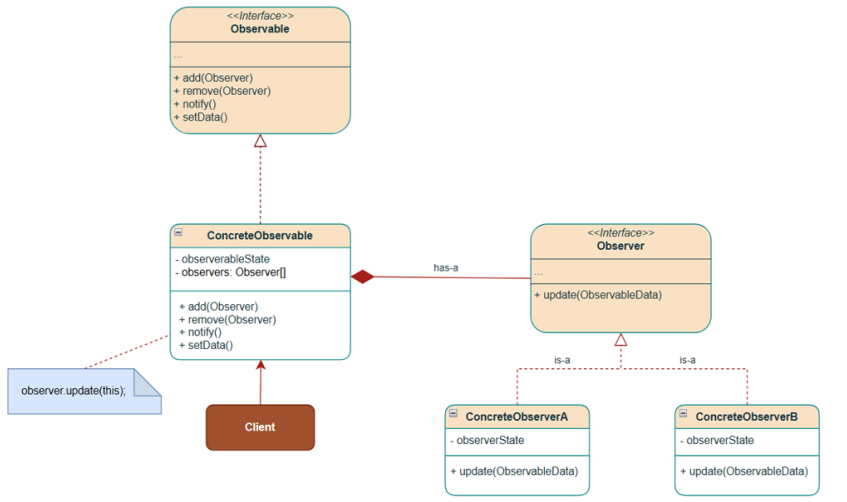


Real-life Examples

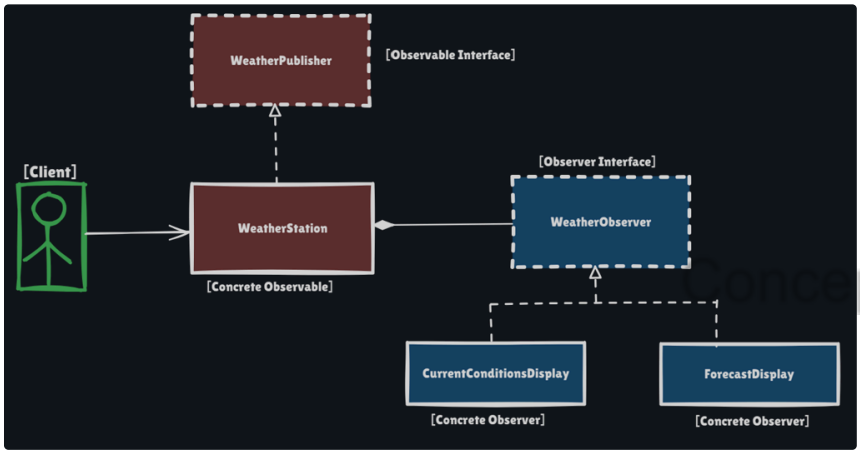
Real-life examples of the Observer pattern include:

- **Weather Applications:** Where multiple devices receive updates from a weather station.
- **Social Media Feeds:** When we follow someone on Instagram, Facebook, or Twitter, we become observer of their profile. When they post new content, we are automatically notified.
- **Subscription Services:** YouTube subscriptions, where viewers are notified of new videos, or Content magazine/newspaper/newsletter subscriptions, where publishers send new issues to subscribers.
- **Stock Market Trackers:** When the price of a stock (or state) changes, the stock's market system (the observable) sends out notifications to all interested investors (the observers).

Class Diagram



Structure of the Observer Pattern



Concept && Coding

Let's understand the Structure of the Observer Pattern using the Weather Station example:

- **Observable Interface (**or Subject Interface, i.e., `WeatherObservable`**)**
  - Defines methods for adding, removing, and notifying observers.
  - The weather station implements this interface.
- **Observer Interface (** `WeatherObserver` **)**
  - Defines the `update()` method that all concrete observers must implement.
  - Called by the observable when there is a change in its state.
- **Concrete Observable (or Concrete Subject, i.e.,** `WeatherStation` **)**
  - Maintains a list of observers.
  - Holds the weather data, i.e., **observable data** (temperature, humidity, pressure).
  - Notifies all observers when measurements(state) change.
- **Concrete Observers (** `ForecastDisplay` **&** `CurrentConditionsDisplay` **)**
  - Each display has different behavior when updated.
  - `CurrentConditionsDisplay` : Shows current weather on gadgets like TV or mobile.

- `ForecastDisplay`: Predicts weather based on pressure changes.

Implementation

1. Example: Weather Station

```java
// Observable(Subject) interface
// Defines methods for managing observers and notifying them of changes
public interface WeatherObservable {

    void addObserver(WeatherObserver observer);

    void removeObserver(WeatherObserver observer);

    void notifyObservers();

    void setWeatherReadings(float temperature, float humidity, float pressure);
}
```

```java
// Concrete Observable (Subject)
// WeatherStation - the concrete observable class that holds weather data
public class WeatherStation implements WeatherObservable {
    // List of observers registered for updates
    private final List<WeatherObserver> observers;
    // Observable Data
    private float temperature;
    private float humidity;
    private float pressure;

    public WeatherStation() {
        observers = new ArrayList<>();
    }

    @Override
    public void addObserver(WeatherObserver observer) {
        observers.add(observer);
        System.out.println("[+] Observer registered: " + observer.getClass().getSimpleName());
    }

    @Override
    public void removeObserver(WeatherObserver observer) {
        observers.remove(observer);
        System.out.println("[-] Observer removed: " + observer.getClass().getSimpleName());
    }

    @Override
    public void notifyObservers() {
        for (WeatherObserver observer : observers) {
            // Notify each observer about the change in weather data(state)
            observer.update(); // Observer will update its state based on the new data and respond accordingly
        }
    }

    // Method to update weather measurements
    public void setWeatherReadings(float temperature, float humidity, float pressure) {
        this.temperature = temperature;
        this.humidity = humidity;
        this.pressure = pressure;
        notifyObservers();
    }
```

Concept && Coding

```java

    // Getters for observers to access weather data
    public float getTemperature() {
        return temperature;
    }

    public float getHumidity() {
        return humidity;
    }

    public float getPressure() {
        return pressure;
    }

    @Override
    public String toString() {
        return "WeatherStation{" +
                "temperature=" + temperature +
                ", humidity=" + humidity +
                ", pressure=" + pressure +
                '}';
    }
}
```

```java
// Observer interface - defines the update method
// Concrete observers implement this interface to update their state
// and respond to changes in its OWN way
public interface WeatherObserver {
    void update();
}
```

```java
// Concrete Observer 1 - Current Conditions Display (on TV or Mobile)
public class CurrentConditionsDisplay implements WeatherObserver {
    private final WeatherObservable weatherStation;

    public CurrentConditionsDisplay(WeatherObservable weatherStation) {
        this.weatherStation = weatherStation;
        weatherStation.addObserver(this);
    }

    // CurrentConditionsDisplay implements the update method in its own way
    @Override
    public void update() {
        System.out.println("Saving weather data... ");
        display();
    }

    // Display the current weather conditions
    public void display() {
        System.out.println("Current Weather Conditions: " + weatherStation.toString());
    }
}
```

Concept && Coding

```java
// Concrete Observer 4 - Forecast Display - Predicts weather based on pressure changes
public class ForecastDisplay implements WeatherObserver {
    private final WeatherObservable weatherStation;

    public ForecastDisplay(WeatherObservable weatherStation) {
        this.weatherStation = weatherStation;
        weatherStation.addObserver(this);
    }

    // ForecastDisplay implements the update method in its own way
    @Override
    public void update() {
```

```java
        System.out.println("Updating weather data to do some analytics: " + weatherStation.toString());
        display();
    }

    // Display the forecast based on the current pressure
    public void display() {
        System.out.println("Forecast Details: Displaying information about Rain, " +
                "Temperature Trends, Significant Weather Events and other phenomemnon...");
    }
}

```

```java
// Client code to demonstrate the Observer Pattern
public class WeatherStationApp {
    public static void main(String[] args) {
        System.out.println("###### State Design Pattern ######");
        // Create the weather station (observable/subject)
        WeatherObservable weatherStation = new WeatherStation();

        // Create displays (observers)
        CurrentConditionsDisplay currentDisplay = new CurrentConditionsDisplay(weatherStation);
        ForecastDisplay forecastDisplay = new ForecastDisplay(weatherStation);

        System.out.println("===>>> Initial Weather Update");
        weatherStation.setWeatherReadings(80, 65, 30.4f);

        System.out.println("===>>> Second Weather Update");
        weatherStation.setWeatherReadings(82, 70, 29.2f);

        // Remove forecast display
        weatherStation.removeObserver(forecastDisplay);

        System.out.println("===>>> Third Weather Update");
        weatherStation.setWeatherReadings(70, 21, 29.2f);
        // Forecast display will not be notified
    }
}
```

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Output

```
###### State Design Pattern ######
[+] Observer registered: CurrentConditionsDisplay
[+] Observer registered: ForecastDisplay
===>>> Initial Weather Update
Saving weather data...
Current Weather Conditions: WeatherStation{temperature=80.0, humidity=65.0, pressure=30.4}
Updating weather data to do some analytics: WeatherStation{temperature=80.0, humidity=65.0, pressure=30.4}
Forecast Details: Displaying information about Rain, Temperature Trends, Significant Weather Events and other phenomemnon...
===>>> Second Weather Update
Saving weather data...
Current Weather Conditions: WeatherStation{temperature=82.0, humidity=70.0, pressure=29.2}
Updating weather data to do some analytics: WeatherStation{temperature=82.0, humidity=70.0, pressure=29.2}
Forecast Details: Displaying information about Rain, Temperature Trends, Significant Weather Events and other phenomemnon...
[-] Observer removed: ForecastDisplay
===>>> Third Weather Update
Saving weather data...
Current Weather Conditions: WeatherStation{temperature=70.0, humidity=21.0, pressure=29.2}

Process finished with exit code 0
```

2. Example: E-commerce “Notify Me” feature

```java
// Observable interface
public interface StockAvailabilityObservable {
    void addStockObserver(StockNotificationObserver observer);

    void removeStockObserver(StockNotificationObserver observer);

    void notifyStockObservers();

```

```java
    boolean purchase(int quantity);

    void restock(int quantity);
}
```

```java
// Concrete Observable
public class IphoneProductObservable implements StockAvailabilityObservable {
    private final String productId;
    private final String productName;
    private final double price;
    private final List<StockNotificationObserver> stockObservers;
    private int stockQuantity;

    public IphoneProductObservable(String productId, String productName, double price, int stockQuantity) {
        this.productId = productId;
        this.productName = productName;
        this.price = price;
        this.stockQuantity = stockQuantity;
        this.stockObservers = new ArrayList<>();
    }

    @Override
    public void addStockObserver(StockNotificationObserver observer) {
        stockObservers.add(observer);
        System.out.println("[+]" + observer.getUserId() + " subscribed for notifications on " + productName);

    }

    @Override
    public void removeStockObserver(StockNotificationObserver observer) {
        stockObservers.remove(observer);
        System.out.println("[-]" + observer.getUserId() + " unsubscribed for notifications on " + productName);
    }

    @Override
    public void notifyStockObservers() {
        if (stockQuantity > 0 && !stockObservers.isEmpty()) {
            System.out.println("Notifying " + stockObservers.size() + " subscribers... ");

            // Create a copy to avoid concurrent modification
            List<StockNotificationObserver> observersToNotify = new ArrayList<>(stockObservers);

            for (StockNotificationObserver observer : observersToNotify) {
                observer.update();
            }
        }
    }

    // Method to restock items
    @Override
    public void restock(int quantity) {
        boolean wasOutOfStock = (stockQuantity == 0);
        stockQuantity += quantity;
        System.out.println("RESTOCKED: " + productName + " - Added " + quantity + " items " + " | " + "Current stock: " + stockQuantity);
        // Only notify if product was previously out of stock
        if (wasOutOfStock && stockQuantity > 0) {
            notifyStockObservers();
        }
    }

    // Method to purchase items
    @Override
```

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```java
    public boolean purchase(int quantity) {
        if (stockQuantity >= quantity) {
            stockQuantity -= quantity;
            System.out.println("PURCHASE SUCCESS: " + quantity + " units of " + productName + " | " + "Remaining stock: " + stockQuantity);
            return true;
        } else {
            System.out.println("PURCHASE FAILED: " + productName + " is out of stock! | " + "Available Quantity: " + stockQuantity);
            return false;
        }
    }

    // Getters
    public String getProductId() {
        return productId;
    }

    public String getProductName() {
        return productName;
    }

    public double getPrice() {
        return price;
    }

    public int getStockQuantity() {
        return stockQuantity;
    }
}

```

```java
// Observer interface for stock availability notifications
public interface StockNotificationObserver {
    void update();

    String getNotificationMethod();

    String getUserId();
}
```

Concept && Coding

```java
// Concrete observer for email notifications
public class EmailNotificationObserver implements StockNotificationObserver {
    private final String userId;
    private final String emailAddress;

    public EmailNotificationObserver(String userId, String emailAddress) {
        this.userId = userId;
        this.emailAddress = emailAddress;
    }

    @Override
    public void update() {
        sendEmail();
    }

    private void sendEmail() {
        System.out.println("!! EMAIL SENT to: " + emailAddress + " - " + "Product is back in stock! Hurry Up!!");
    }

    @Override
    public String getNotificationMethod() {
        return "Email";
    }

    @Override
    public String getUserId() {
```

```
        return userId;
    }
}
```

```
// Concrete observer for push notifications
public class PushNotificationObserver implements StockNotificationObserver {
    private final String userId;
    private final String deviceToken;

    public PushNotificationObserver(String userId, String deviceToken) {
        this.userId = userId;
        this.deviceToken = deviceToken;
    }

    @Override
    public void update() {
        sendPushNotification();
    }

    private void sendPushNotification() {
        System.out.println("!! PUSH NOTIFICATION SENT to: " + deviceToken + " - " + "Product is back in stock! Hurry Up!!");
    }

    @Override
    public String getNotificationMethod() {
        return "Push Notification";
    }

    @Override
    public String getUserId() {
        return userId;
    }
}
```

Concept && Coding

```
public class ECommerceStockNotificationApp {
        System.out.println("----------------------------------------------------------------------------------");
        System.out.println("###### E-commerce Store - Stock Availability Notification Feature Demo ######");

        // Create an iPhone product - stock available = 10 units
        StockAvailabilityObservable iphoneProduct = new IphoneProductObservable("ip15", "iphone 15", 1250, 10);

        // Create observers
        StockNotificationObserver John_PUSH = new PushNotificationObserver("John123", "JohnDeviceP1");
        StockNotificationObserver Katy_PUSH = new PushNotificationObserver("Katy678", "KatyDeviceP2");
        StockNotificationObserver Jane_EMAIL = new EmailNotificationObserver("Jane783", "jane783@gmail.com");
        StockNotificationObserver George_EMAIL = new EmailNotificationObserver("George993", "george993@gmail.com");

        // Black Friday Sale - Purchase all 10 units
        iphoneProduct.purchase(10);

        // Stock unavailability leads to users subscribing to notifications
        boolean success = iphoneProduct.purchase(1); // Failed purchase
        if (!success) {
            // Register observers - John, Katy, Jane, George subscribe for notifications upon stock availability
            iphoneProduct.addStockObserver(John_PUSH); // John
            iphoneProduct.addStockObserver(Katy_PUSH); // Katy
            iphoneProduct.addStockObserver(Jane_EMAIL); // Jane
            iphoneProduct.addStockObserver(George_EMAIL); // George
```

```java
        }

        // Restock 20 units of iPhone 15
        iphoneProduct.restock(20); // All 4 observers are notified

        // Users purchase upon receiving notifications
        iphoneProduct.purchase(1); // John purchases 1 unit
        iphoneProduct.purchase(1); // Katy purchases 1 unit

        // John & Katy unsubscribe from notifications
        iphoneProduct.removeStockObserver(John_PUSH);
        iphoneProduct.removeStockObserver(Katy_PUSH);

        // NYE Sale - All 18 units sold
        iphoneProduct.purchase(18);
        iphoneProduct.restock(5); // Only Jane & George are notified

        iphoneProduct.purchase(1); // Jane purchases 1 unit
        iphoneProduct.purchase(1); // George purchases 1 unit

        // Jane & George unsubscribe from notifications
        iphoneProduct.removeStockObserver(Jane_EMAIL);
        iphoneProduct.removeStockObserver(George_EMAIL);
    }
}
```

Output

```
--------------------------------------------------------------------------
###### E-commerce Store - Stock Availability Notification Feature Demo ######
PURCHASE SUCCESS: 10 units of iphone 15 | Remaining stock: 0
PURCHASE FAILED: iphone 15 is out of stock! | Available Quantity: 0
[+]John123 subscribed for notifications on iphone 15
[+]Katy678 subscribed for notifications on iphone 15
[+]Jane783 subscribed for notifications on iphone 15
[+]George993 subscribed for notifications on iphone 15
RESTOCKED: iphone 15 - Added 20 items  | Current stock: 20
Notifying 4 subscribers...
!! PUSH NOTIFICATION SENT to: JohnDeviceP1 - Product is back in stock! Hurry Up!!
!! PUSH NOTIFICATION SENT to: KatyDeviceP2 - Product is back in stock! Hurry Up!!
!! EMAIL SENT to: jane783@gmail.com - Product is back in stock! Hurry Up!!
!! EMAIL SENT to: george993@gmail.com - Product is back in stock! Hurry Up!!
PURCHASE SUCCESS: 1 units of iphone 15 | Remaining stock: 19
PURCHASE SUCCESS: 1 units of iphone 15 | Remaining stock: 18
[-]John123 unsubscribed for notifications on iphone 15
[-]Katy678 unsubscribed for notifications on iphone 15
PURCHASE SUCCESS: 18 units of iphone 15 | Remaining stock: 0
RESTOCKED: iphone 15 - Added 5 items  | Current stock: 5
Notifying 2 subscribers...
!! EMAIL SENT to: jane783@gmail.com - Product is back in stock! Hurry Up!!
!! EMAIL SENT to: george993@gmail.com - Product is back in stock! Hurry Up!!
PURCHASE SUCCESS: 1 units of iphone 15 | Remaining stock: 4
PURCHASE SUCCESS: 1 units of iphone 15 | Remaining stock: 3
[-]Jane783 unsubscribed for notifications on iphone 15
[-]George993 unsubscribed for notifications on iphone 15
```

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